National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

CS 1005 : Discrete Structures

Assignment 1

Instructor : Ms Mahzaib Younas

Section C

Q No 01: Find the negation of the proposition "Vandana's smartphone has at least 32 GB of memory" and express this in simple English and in expression.

"It is not the case that Vandana's smartphone has at least 32 GB of memory."
This negation can also be expressed as
"Vandana's smartphone does not have at least 32 GB of memory"

Q No 2: Find the conjunction of the propositions p and q (p ^ q), where p is the proposition "Rebecca's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

p is the proposition "Rebecca's PC has more than 16 GB free hard disk space"

q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

"Rebecca's PC has more than 16 GB free hard disk space, and its processor runs faster than 1 GHz."

For this conjunction to be true, both conditions given must be true. It is false when one or both of these conditions are false.

Q No 03: What is the disjunction of the propositions p and q, where p is the proposition "Rebecca's PC has more than 16 GB free hard disk space" and q is the proposition "The processor in Rebecca's PC runs faster than 1 GHz."

The disjunction of p and q, $p \vee q$, is the proposition Extra Examples
"Either Rebecca's PC has at least 16 GB free hard disk space, or the processor in Rebecca's PC runs faster than 1 GHz."

Q No 04: Prove the logarithm Quotient law, if b=5, x=75 and y=125 respectively.

The logarithm Quotient law is

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

LHS	RHS
$=\log_b\left(\frac{x}{y}\right)$ $\log_5\left(\frac{75}{125}\right)$ $\log_5\left(\frac{3}{5}\right)$ -0.31739	$\log_b x - \log_b y$ $\log_5 75 - \log_5 125$ $2.68261 - 3$ -0.31739

Q No 05: Describe the rules of addition with appropriate example?

There are four rules of additions

Additive Inverse	**Commutative**	**Associativity**	**Cancellation rule**
General eq **a + (-a) = 0** let a = 5 then its additive inverse will be -5 5 + (-5) = 0 0 = 0 LHS = RHS	**a + b = b + a** let suppose a=2 and b=7 2+7 =7+2 **9 = 9** **LHS= RHS**	**(a + b) + c = a + (b + c)** let suppose a=2, b=7, and c=10 **(2+7)+10 = 2 + (7+10)** **(9) + 10 = 2 + (17)** **19 = 19** **LHS = RHS**	If a + b = a + c, then b = c a + b = a + c taking additive inverse of a which is (-a) and add on both sides a + b +(-a) = a + c + (-a) **b=c**

Q No 06: Make a truth table for the statement

1. **(P ∨ Q) ∧ ~(P ∧ Q)**

P	**Q**	**(P ∨ Q)**	**(P ∧ Q)**	**~(P ∧ Q)**	**(P ∨ Q) ∧ ~(P ∧ Q)**
T	**T**	**T**	**T**	**F**	**F**
T	**F**	**T**	**F**	**T**	**T**
F	**T**	**T**	**F**	**T**	**T**
F	**F**	**F**	**F**	**T**	**F**

2. $(P \lor Q) \rightarrow (P \land Q)$

P	Q	(P ∨ Q)	(P ∧ Q)	(P ∨ Q) → (P ∧ Q)
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

Q No 07: Make a truth table and prove that given condition

$$\neg (P \lor (\neg P \land Q)) = (\neg P \land \neg Q)$$

P	Q	¬P	(¬P ∧ Q)	(P ∨ (¬P ∧ Q))	LHS	RHS
					¬ (P ∨ (¬P ∧ Q))	(¬P ∧ ¬Q)
T	T	F	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	F	T	F	F	T	T